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Calibration of Vehicle Following Model Parameters Using Mixed Traffic Trajectory Data

Abstract

A number of models for car following have been proposed for homogeneous traffic and some of these have been modified or adapted to represent mixed traffic conditions in India. Vehicle following behaviour under mixed traffic is both complex and challenging and cannot be adequately captured by conventional lane-based following models and their variants. For example, the behavior of a subject vehicle in mixed traffic condition depends on the behavior of lead vehicle as well as the influence of neighboring vehicles. Most existing models are based on the longitudinal spacing and the relative speed of the lead and the subject vehicles. However, the vehicular interactions also depend on the lateral movements such as lateral spacing and lateral speed. Further, the response of the subject vehicle also depends on the type of vehicles involved and their maneuvers in surrounding space. This study aims to address some of these gaps in existing vehicle following models for mixed traffic. Mixed traffic trajectory data collected from the mid-block section of a six-lane divided urban arterial road in Chennai city was used for this study. From the data set, leader-follower vehicle pairs identified based on three different methods: influence area method, headway method and video data tracking are compared and more suitable method is chosen for further analysis. Variation of driver behavior due to different factors such as follower's speed, relative spacing, lateral position, vehicle types and following behaviors were examined. Calibration and validation of the models was done for different leader-follower vehicle types. The results show that the model parameters vary with not only subject vehicle type but also leader-follower pairs. Also, there is a significant effect of factors like lateral position of vehicles and types of following behaviors. This study will find application in developing more realistic mixed traffic simulation models by including these factors.

Keywords: Leader-follower pairs, Influence area, Vehicle following model, Calibration, Mixed traffic, Trajectory dataset

1. Introduction

Studies on car following models have been adapted to represent the real traffic conditions, especially in mixed traffic conditions. For lane based conditions Pipes [1] presented a car following model concept based on safe distance and Gazis-Herman-Rothery (GHR) proposed a family of models based on stimulus-response concept which is probably the most studied model class [2]. After this many concepts regarding following behavior have been introduced. These include collision avoidance model based on the safe-following distance that driver tries to keep to avoid collision [3-5]; and psycho-physical models based on the concept that driver follows with different regimes based on some threshold [6-7]; and optimum velocity model based on the assumption that the driver accelerate or decelerate based on his/her speed and the optimum speed that he/she can attain [8].

Most of the car-following studies were developed for homogeneous traffic conditions. Kesting and Trieber [9] performed calibration of intelligent driver model and speed difference model using the NGSIM trajectory dataset and found that intelligent driver model is more robust. Chen and Li [10] evaluated the Intelligent Driver Model (IDM) and the Microscopic Traffic SIMulator (MITSIM) car-following model and found that neither of them can simulate all the driving behaviors, but both provide an acceptable approximation of observed driving patterns. Punzo and Tripodi [11] developed a

different approach to calibrate Gipps microscopic traffic flow model, where in the values of calibrated parameters were all significant and consistent with expectations.

In mixed traffic conditions, most of the models were adapted or derived from conventional car-following models. Toledo *et al.* [12] studied the calibration and validation approach by applying a microscopic traffic simulation tool MITSIMLab to a mixed urban-freeway network. Gunay [13] has developed a car-following model by taking lateral discomfort into consideration and introduced a new concept for the lateral behavior of follower vehicle in car-following model. Ravishankar and Mathew [14] modified the Gipps car-following model to incorporate vehicle type dependent parameters and found that the modification improved the prediction of following behavior. Asaithambi and Basheer [15] also developed a model which found that the following behaviors are different for different types of vehicles. Durrani *et al.* [16] calibrated the Weidemann's vehicle-following models based on vehicle-pair interactions. It was found that heavy vehicles maintain longer spacing and time gap with the lead vehicle and are less sensitive to the lead car behavior. Asaithambi *et al.* [17] evaluated different car-following models under mixed traffic conditions. From the selected models it was found that the Gipps and IDM models give reasonable critical parameter values. Raju *et al.* [18] calibrated vehicle-following model under heterogeneous traffic condition using trajectory data from two study sections. The simulation models calibrated with field data better replicated the real world condition compared to default parameters.

Vehicle following behaviors under mixed traffic is both complex and challenging and cannot be adequately captured by conventional lane-based following models and their variants. Most existing models are based on the longitudinal spacing and the relative speed of the lead and the subject vehicles. However the vehicular interactions also depend on the lateral behaviors such as lateral spacing and lateral speed. Further, the response of the subject vehicle also depends on the type of vehicles involved and their maneuvers in mixed traffic. This study aims to address some of these gaps in existing vehicle-following models for mixed traffic. The study aims to investigate vehicle following behaviour under mixed traffic by focusing on three related objectives:

1. To compare different methods for the identification of leader-follower pairs
2. To analyze and understand the effect of various factors on following behavior
3. To estimate suitable vehicle-following models for different types of leader-follower pairs

To achieve these objectives, trajectory data from a study site in Chennai was used. To investigate the first objective, leader-follower pairs were identified based on influence area method, headway method and from video tracking. Among the three methods, the better method was identified and selected leader-follower pairs were used for further analysis. For achieving the second objective, following behavior was investigated under varying factors by comparing the corresponding sensitivity values. For this purpose, the following factors and levels were considered: follower's speed (low, medium and high ranges), relative longitudinal spacing (low, medium and high ranges), vehicles on different lanes (lane 1, lane 2 and lane 3), vehicle following behavior (strict, staggered and non-overlapping following) and different leader-follower vehicle types. For the selected leader-follower pairs the GM model was estimated using the entire data and the sensitivity coefficient under different factors was studied. For the third objective, two vehicle following models were calibrated: Gipps and GM at both a disaggregate level

(vehicle pair type-wise) and aggregate level (same model for all types). Six leader-follower vehicle type combinations considered in this study were car- two wheeler, car-car, two wheeler-two wheeler, two wheeler-auto and auto –two wheeler. After calibration, validation of the models was done by comparing positions and speeds.

2. Data Description

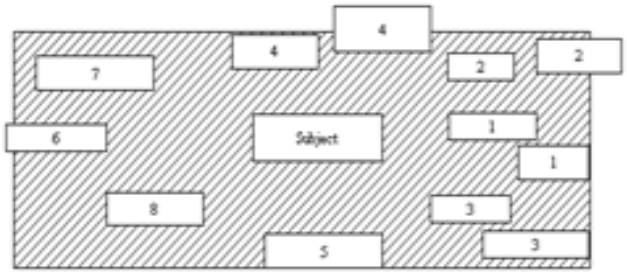
Videographic data were collected from a six-lane divided urban arterial road at the Maraimalai Adigalar Bridge in Saidapet, Chennai, India for this study [19]. The selected section is on a bridge which ensures the uniformity of section. The video data were recorded from 10:00 AM to 3:30 PM in February 2014. The coordinates, dimensions and class of all vehicles in the video sequences during a period of 30 minutes between 2:45 PM and 3:15 PM were obtained using a trajectory extractor. The trajectories were extracted at a time resolution of 0.5 second. Traffic composition at this study stretch comprises 56.4% two-wheelers (TW), 26.6% passenger cars (car), 12.2% auto-rickshaws (Auto), 3.2% buses and 1.6% including light and heavy trucks.

3. Identification of Leader-Follower Pairs

Three different methods were considered for identifying leader-follower pairs and their efficiency in identification is analyzed. A sample of 300 leader-follower pairs was identified through each method.

3.1 Method Based on Influence Area Concept

Influence area may be defined as the area which contains the most influencing surrounding vehicles (which affects the subject vehicle). Figure 1 illustrates the influence area bounded by a subject vehicle. The influence area is initially assumed to be a rectangle with maximum front and rear boundary as 30 m (in VISSIM, look ahead distance for urban roads is 30m [20]) and maximum left and right side boundary as 3 m (considered approximately a lane width). Next the left side adjacent vehicles which are longitudinally overlapping with the subject vehicle within the maximum lateral boundary are identified. Then the left boundary is taken as the left edge of the nearest adjacent vehicle on the left which is within the tentative left boundary above. The right side boundary is defined similarly on the right hand side. For front boundary, vehicles in front of the subject vehicle, which are laterally overlapping with the subject vehicle and within the maximum longitudinal boundary are identified. The front boundary is taken as the farthest front edge of leading vehicles contained within the front of the initial rectangle above. If laterally overlapping leader is not present, the non-overlapping vehicles within the left and right boundary are considered. The rear boundary is similarly defined, considering vehicles which are longitudinally behind the subject vehicle within the tentative rectangle above. If no vehicles are present in a particular side (front, rear, left or right), then the tentative boundary is fixed as the boundary of the influence area on that side.



- Position 1 – Leaders
- Position 2 – Left side non-overlapping leaders
- Position 3 – Right side non-overlapping leaders
- Position 4 – Left side adjacent vehicles
- Position 5 – Right side adjacent vehicles
- Position 6 – Followers
- Position 7 – Left side non-overlapping follower
- Position 8 – Right side non-overlapping follower

Figure 1 Influence area around a subject vehicle.

3.2 Method Based on Headway Concept

In this method, leader-follower pairs were identified based on time headway and lateral overlapping criterion. The time headway limit is considered as two seconds in this study. For a given leader, consecutive vehicles that are laterally overlapping with the leader and pass through the same section as the leader within the headway limit above, are considered as strictly following the lead vehicle. From the identified follower pairs, the follower closer to the leader is considered as the real follower.

3.3 Method Based on Leader-Follower Pairs from Video Tracking

In this method, leader-follower pairs were identified based on visual tracking of vehicles from the video data by noting the vehicles movement behind a given vehicle (within a reasonable lateral gap) which responded to change in speed, longitudinal position or lateral position of lead vehicle.

3.4 Analysis of Leader-Follower Pairs by Different Methods

From the three methods, video method is taken as the reference method since data from multiple frames were used and the following pattern was ascertained manually. The confusion matrix (Table 1), accuracy, misclassification rate, recall, false positive rate, specify and precision were found out and the results are shown in Table 2. Accuracy and recall is higher for the influence area method which shows that influence area method is better compared to the headway method to predict the following pairs. Precision value is higher for headway method. Results show that influence area method identifies more leader-follower pairs than the headway method. Both methods have some percentage of error compared to video tracking method. The error for influence area method and headway method are 20% and 36% respectively. This can yield differences in following models and hence comparison was done by developing GM models below in section 3.5.

Table 1 Confusion matrix.

Actual method \ Predicted method		Influence area method			Headway method		
		Following (YES)	Not following (NO)	Total	Following (YES)	Not following (NO)	Total
Video method	Following (YES)	192 (62.74%) TP	38 (12.42%) FN (Type II error)	230	125 (40.85%) TP	105 (34.31%) FN (Type II error)	230
	Not following (NO)	24 (7.84%) FP (Type I error)	52 (17%) TN	76	4 (1.31%) FP (Type I error)	72 (23.53%) TN	76
	Total	216	90	306	129	177	306

Legend: TP – True Positive, FN – False Negative, FP – False Positive, TN – True Negative.

Table 2 Results obtained from confusion matrices.

Output	Influence Area method	Headway method
Accuracy	0.80	0.64
Misclassification Rate	0.2	0.36
True Positive Rate /Sensitivity/Recall	0.83	0.54
False Positive Rate	0.32	0.05
Specify	0.68	0.95
Precision	0.89	0.97

3.5 Comparison of GM Model across Different Leader-Follower Identification Methods

The first General Motors model was considered in this study and the model formulation is shown in the following equation.

$$a\,(t+T) = \lambda * dv(t) + \varepsilon \qquad (1)$$

Where, a represents the acceleration of follower vehicle, λ is the sensitivity, dv is the relative speed at time t and ε represents intercept.

The regression analysis was done for different methods and the results are shown in Table 3. Magnitudes of intercept for the three methods are similar but insignificant for headway and video data method at 5% level. Sensitivities do not vary much in magnitude or significance. However, goodness of fit is best for video data method and less for the other two methods.

Table 3 Comparison of Leader-Follower identification methods.

Statistic	Video Method	Influence Area Method	Headway Method
Correlation coefficient	0.79	0.75	0.77
R^2	0.62	0.59	0.59

Standard error	0.53	0.59	0.48
F stat	0.00	0.00	0.00
Sensitivity	0.40	0.39	0.38
P-value sensitivity	0.00	0.00	0.00
Intercept	0.01	0.02	0.013
P-value intercept	0.09	0.04	0.37

4. Effect of Various Factors on Driver Behavior

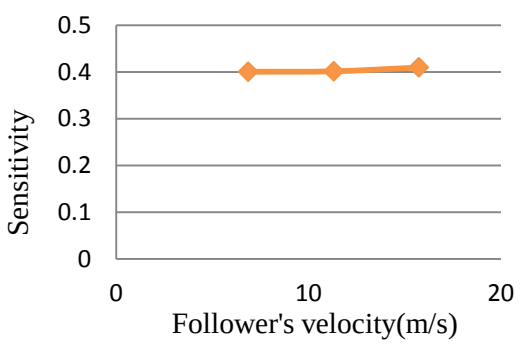
Driving behavior at different speed and spacing ranges, and for different vehicle types, lateral position and following behaviours are analyzed by comparing variation of sensitivity. Using the distribution curve, low, medium and high ranges of speed and positions were obtained. The low, medium and high speed ranges were considered are 3.3-10.43 m/s, 10.43-12.19 m/s and 12.19-19.26 m/s, respectively. The range for low, medium and high longitudinal spacing are 0.01-8.12 m, 8.12-13.29 m and 13.29-42.95 m, respectively. Lateral positions were analyzed by comparing leader-follower pairs in different lanes into consideration. Car-two-wheeler, car-car, car-auto-rickshaw, two-wheeler-two-wheeler, two-wheeler-car, two-auto-rickshaw, auto-rickshaw-car and auto-rickshaw-two-wheeler are the vehicle pair types considered in this study.

4.1 Variation of Sensitivity across Follower's Speed

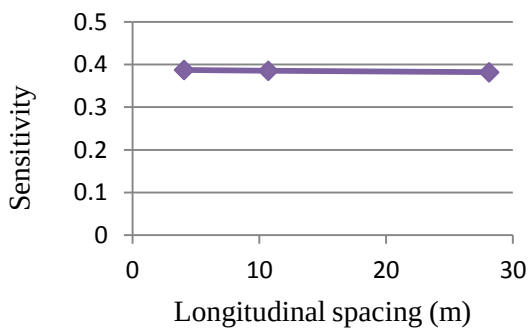
Variation of sensitivity across the speed ranges are shown in Figure 2 (a). Sensitivity across different speed ranges do not appear to vary significantly. However, this can be due to the vehicle heterogeneity. As majority of the vehicle composition consists of two-wheelers, which may show similar sensitivity for all speed ranges. However, the trend is increasing, the follower will be more cautious at higher speed.

4.2 Variation of Sensitivity across Longitudinal Spacing

Figure 2 (b) shows the variation of sensitivity as longitudinal spacing increases. Again there is no significant difference between sensitivity for different spacing ranges. Spacing between leader-follower does not appear to have much significant impact on driver behavior. Graph shows a decreasing trend at higher spacing.



(a) Follower's speed.



(b) Longitudinal spacing.

Figure 2 Variation of sensitivity across follower's speed and longitudinal spacing.

4.3 Variation of Sensitivity across Lanes

The lane near to the shoulder is considered as lane 1, the middle lane as lane 2 and the lane near to median as lane 3. Variation of sensitivity across lane is shown in Fig 3 (a). Vehicle pairs in lane 1 shows a larger sensitivity than vehicle pairs in lane 2 and lane 3. It is observed that in lane 1 the composition mainly consists of two-wheelers and auto-rickshaws; wherein the other vehicles are observed to follow with more care. Also the presence of footpath and pedestrians near lane 1 increases the sensitivity. The presence of median in lane 3 also makes the sensitivity in lane 3 slightly more than lane 2.

4.4 Variation of Sensitivity across Vehicle Types

Table 4 shows the variation in sensitivity across leader-follower vehicle types. When car is the lead vehicle, auto-rickshaw shows higher sensitivity while following than car and two-wheeler (as followers). It was found that when an auto-rickshaw follows another vehicle, acceleration regimes and smaller longitudinal spacing are more commonly observed. From video also it is observed that while following another vehicle, auto-rickshaw has a higher tendency to overtake. When two-wheeler is the leader, car shows more sensitivity while following than two-wheeler and auto-rickshaw possibly due to larger size and higher speed. When auto-rickshaw is the leader, two-wheeler shows larger sensitivity. As two-wheeler has better manoeuverability they show a tendency to overtake slow moving auto-rickshaws.

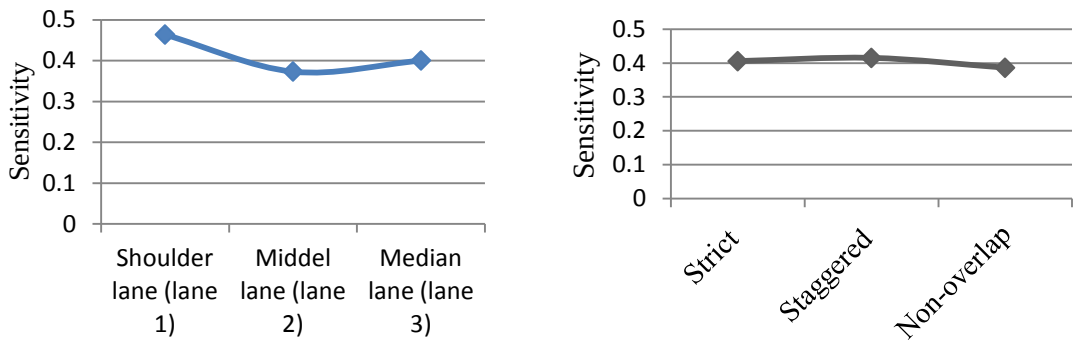
Table 4 Variation of sensitivity across vehicle types.

Leader-follower pairs	Car-TW	Car-Car	Car-Auto	TW-TW	TW-Car	TW-Auto	Auto - TW	Auto - Car	Auto - Auto
Sensitivity	0.37	0.39	0.50	0.41	0.43	0.38	0.45	0.34	0.43

4.5 Variation of Sensitivity across Following Behavior

Figure 3 (b) shows variation of sensitivity across different types of following behaviours. If the centre line difference between leader and follower is less than or equal to 0.5 m it is classified as strict following. If centre line difference is more than 0.5 m, and also, lateral overlapping is there, it is classified as staggered following. If lateral overlapping is not there, it is classified as following without lateral overlapping category.

From strict following to non-overlapping following, sensitivity decreases. Follower may be more cautious when he/she follows a leader with lateral overlapping as collision probability is more in following with lateral overlap case compared to non-overlapping following case. Sensitivity of staggered following is high compared to strict and non overlapping following. In staggered following more chance is there to overtake the vehicles in both sides, this causes the follower to be more cautious.



(a) Across lanes. **(b) Across following behaviors.**

Figure 3 Variation of sensitivity across lanes and following behaviors.

5. Calibration of Vehicle-Following Models

5.1 Vehicle-Following Models

Two different models are calibrated and separate parameters are estimated for the entire data and also for each vehicle pair type. The best model for each is identified by comparing goodness of fit of models.

5.1.1 General Motors model

This model is developed by Gazis, Herman and Rothery [2] based on the concept that the follower will accelerate or decelerate based on the stimulus he or she received from the leading vehicle. Based on this concept, a linear model was built between the stimulus and response. Updated acceleration of follower at time t+T is represented in equation 1

5.1.2 Gipps Model

Gipps [3] formulated a new model based on the philosophy that each driver sets limits to his/her desired braking and acceleration rates. The parameters in this model correspond to the characteristics of driving behavior. The updated speed at time t+T based on free-flow conditions and car-following conditions are represented in equation 2 and 3, respectively.

$$v^a_{t+T_i} = v^f_T + 2.5 a_j T_j \left(1 - \frac{v^f_t}{V_j}\right) \sqrt{0.025 + \frac{v^f_t}{V_j}} \qquad (2)$$

$$v^b_{t+T_j} = b_{ij} T_j + \sqrt{b^2_{ij} T_j^2 - b\left[2\left(x^l_t - x^f_t - S_i\right) - v^f_t T_j - \frac{(v^l_t)^2}{b^*_{ij}}\right]} \qquad (3)$$

Where, i and j represent lead and following vehicle, respectively; v^a_{t+Tj} and v^b_{t+Tj} are the updated speed of follower vehicle at time t+T for acceleration and deceleration conditions, respectively; T_j is the reaction time, b_{ij}, and b^*_{ij} are deceleration of the following vehicle and assumed deceleration of lead vehicle, respectively; S_i is the effective size of lead vehicle I; a_j is the maximum acceleration of following vehicle j; V_j

is the desired speed of following vehicle j,; x_t^l and x_t^f are the leader's and follower's position at time t, respectively and v^f_t and v^l_t are leader's and follower's speed at time t, respectively.

5.2 Model Calibration

Based on the equations, optimized parameter values were calculated using Solver tool in Microsoft Excel. Calibrated parameter values for both Gipps and GM model for the aggregate data and for different leader-follower vehicle type combinations are shown in Table 5. For car-car and car-two-wheeler pairs, Gipps model fits better compared to GM model and for two-wheeler-two-wheeler, two-wheeler-car, two-wheeler-auto-rickshaw and auto-rickshaw-two-wheeler pairs, GM model fits better. The results justify the need of a hybrid model in mixed traffic conditions. This model will be combination of two or more models for different leader-follower vehicle types.

Model is corroborated to check the consistency of the models with positions and speeds on trajectory data using three measures: Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE) and t-test for speeds and positions. Validation results were compared with old parameters values (Kanagaraj *et al.* [19]) using t-test, MAPE and RMS as shown in Table 6. It is observed that means of speed and position of observed and model are not significantly different at 0.05 level of significance. MAPE is lesser than 15% for all leader-follower combinations. The t-test, MAPE and RMSE values of position obtained for old and calibrated parameters are shown in Table 7. Figures 4 and 5 shows the trajectory of followers to compare the observed longitudinal position and longitudinal position (obtained from calibrated parameter values) for Gipps and GM model respectively.

Table 5 Calibrated parameters of Gipps and GM model for aggregate and leader-follower categories.

Parameters		Aggregate data	Car-Car	TW-TW	Car-TW	TW-Car	TW-Auto	Auto-TW
Gipps model	a (m/s^2)	2.67	1.81	4.67	2.32	1.78	2.44	2.13
	V (m/s)	11.21	11.54	13.17	12.20	11.78	10.82	10.03
	b (m/s^2)	-2.46	-1.97	-3.13	-2.69	-0.62	-1.08	-1.09
	b* (m/s^2)	-2.24	-1.91	-3.26	-2.26	-0.42	-0.44	-0.44
	S (m)	3.83	5.18	0.50	3.68	3.95	3.82	3.82
	T(s)	0.25	0.21	0.24	0.35	0.59	1.47	0.98
	R^2	**0.71**	**0.82**	0.59	**0.72**	0.62	0.04	0.24
GM model	R^2	0.62	0.61	**0.67**	0.64	**0.72**	**0.61**	**0.71**
	F stat	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Sensitivity	0.40	0.39	0.41	0.37	0.43	0.38	0.46
	P-value sensitivity	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Intercept	0.01	-0.04	-0.02	0.01	-0.02	-0.10	0.09
	P-value intercept	0.09	0.01	0.467	0.81	0.62	0.22	0.03

Table 6 Comparison of Model and observed speed using t-test, MAPE and RMSE.

Leader-Follower	Gipps model	GM model	t-test				MAPE and RMSE			
			Default parameters (Gipps model)		Calibrated parameters		Default parameters (Gipps model)		Calibrated parameters	
			t-stat	t-critical	t-stat	t-critical	MAPE	RMSE	MAPE	RMSE
Car-TW	✓		4.60	±1.97	0.72	1.97	8.52	1.50	7.32	1.30
Car-Car	✓		9.41	±1.97	0.88	1.97	28.08	3.91	7.29	1.41
TW-TW		✓	9.57	±1.98	0.82	1.98	21.80	3.07	14.06	2.28
TW-Car		✓	7.60	±1.98	-1.75	1.98	34.22	4.00	14.32	1.84
TW-Auto		✓	3.86	±1.98	-0.67	±1.98	13.03	1.95	12.8	1.64
Auto-TW		✓	1.96	±1.97	1.74	±1.97	9.33	1.16	11.70	1.47

Table 7 Comparison of Model and observed position using t-test, MAPE and RMSE.

Leader-Follower	Gipps model	GM model	t-test				MAPE and RMSE			
			Default parameters (Gipps model)		Calibrated parameters		Default parameters (Gipps model)		Calibrated parameters	
			t-stat	t-critical	t-stat	t-critical	MAPE	RMSE	MAPE	RMSE
Car-TW	✓		1.16	1.97	0.53	±1.97	5.75	7.97	3.04	4.16
Car-Car	✓		2.82	1.97	0.78	±1.98	12.37	20.82	3.94	6.54
TW-TW		✓	1.41	1.98	0.88	±1.98	6.73	13.35	4.82	9.94
TW-Car		✓	2.53	1.98	-0.12	±1.98	10.95	19.28	3.17	4.56
TW-Auto		✓	1.09	1.98	-0.13	±1.98	4.51	10.38	1.41	2.99
Auto-TW		✓	0.29	1.97	0.17	±1.97	1.95	2.98	2.21	4.17

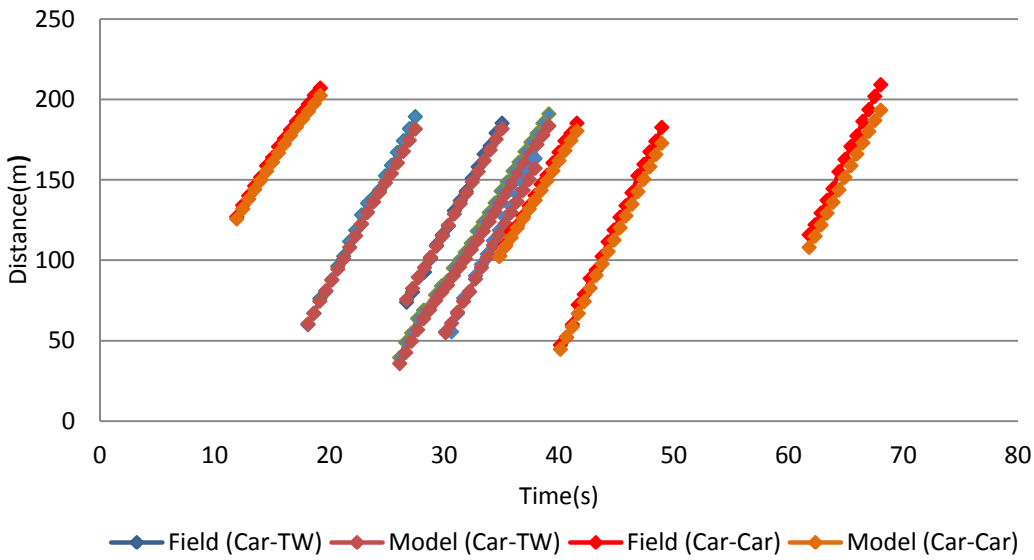


Figure 4 Follower trajectories for field and observed data using Gipps model.

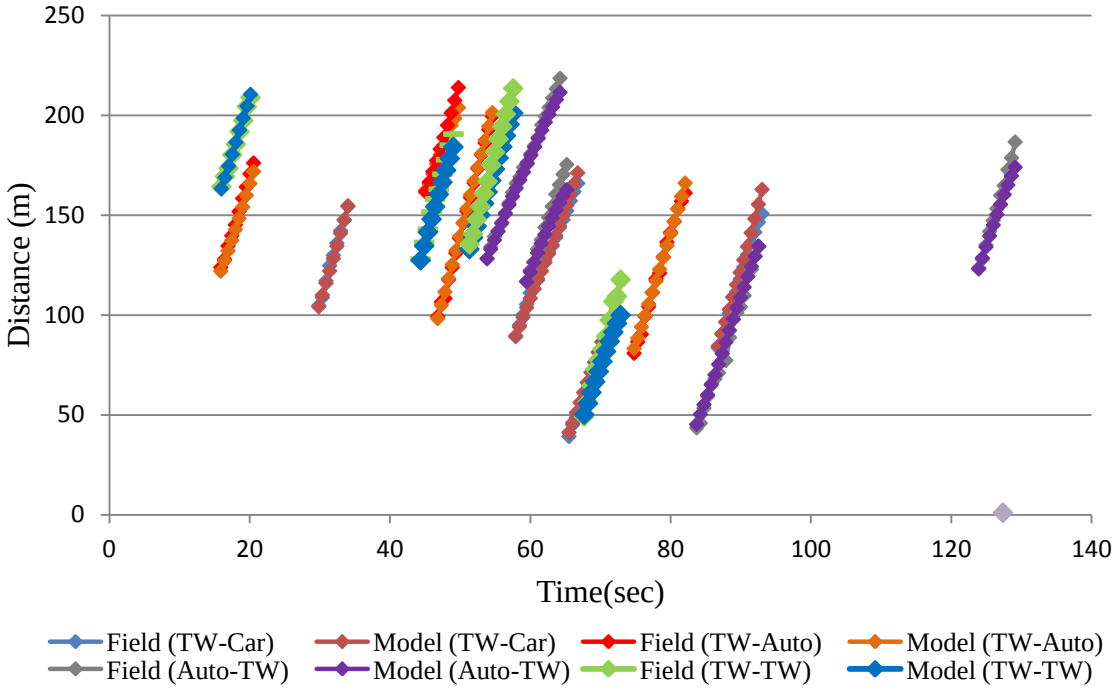


Figure 5 Follower trajectories for field and observed data using GM model.

Ability of selected models to represent non-following pairs were analyzed and represented in Tables 8 (a) and 8 (b). Comparison between aggregate following and non following leader-follower pairs for both GM and Gipps model was presented in the tables. The performance of GM model is poor compared to Gipps model.

Table 8 Calibration of non-following Leader-Follower (L-F) pairs using GM and Gipps model.

(a) GM model parameters.

Output Summary	Following L-F pairs	Non-Following L-F pairs
Correlation coefficient	0.79	0.67
R^2	0.62	0.44
F stat	0.00	0.00
Sensitivity	0.40	0.12
P-value sensitivity	0.00	0.00
Intercept	0.01	-0.05
P-value intercept	0.09	0.01

(b) Gipps model parameters.

Parameters	Following L-F pairs	Non-Following L-F pairs
R^2	0.71	0.91
a (m/s^2)	2.67	0.15
V (m/s)	11.21	9.10
b (m/s^2)	-2.46	-3.36
b* (m/s^2)	-2.24	-0.93
S (m)	3.83	3.72
T(s)	0.25	0.73

6. Conclusions

This paper focused on the identification and analysis of leader-follower vehicle pairs in mixed traffic conditions using mixed traffic trajectory data. Selected leader-follower pairs were calibrated and model parameters for Gipps and General Motors (GM) models for different leader-follower vehicle types were obtained. The key conclusions of this study are:

1. Among the three methods considered, video method was found to be more suitable to determine leader-follower pairs. Method based on influence area concept is more accurate than the method based on headway concept.
2. Following behavior is affected by factors like follower's speed, relative spacing, lateral position, vehicle type and following behavior. It is essential to incorporate all these parameters in the model to get a more accurate model.
3. Calibration of vehicle-following models shows that the Gipps model was more suitable for car-car, car-two-wheeler and GM model was suitable for two-wheeler-two-wheeler, two-wheeler-car, two-wheeler-auto and auto- two-wheeler pairs.
4. The results suggest that different leader-follower pair data are fitted better by different models, which imply the need for hybrid model, which is a combination of two or more models, instead of a single model.
5. Validation results show that the models replicate the field condition satisfactorily.

This study will find application for developing a simulation model, which can provide a more accurate representation of real traffic conditions prevailing on urban roads in India. Further, consideration of models based on the vehicle types in mixed traffic conditions will lead to improvements in simulation model.

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